



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO PRANAV AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

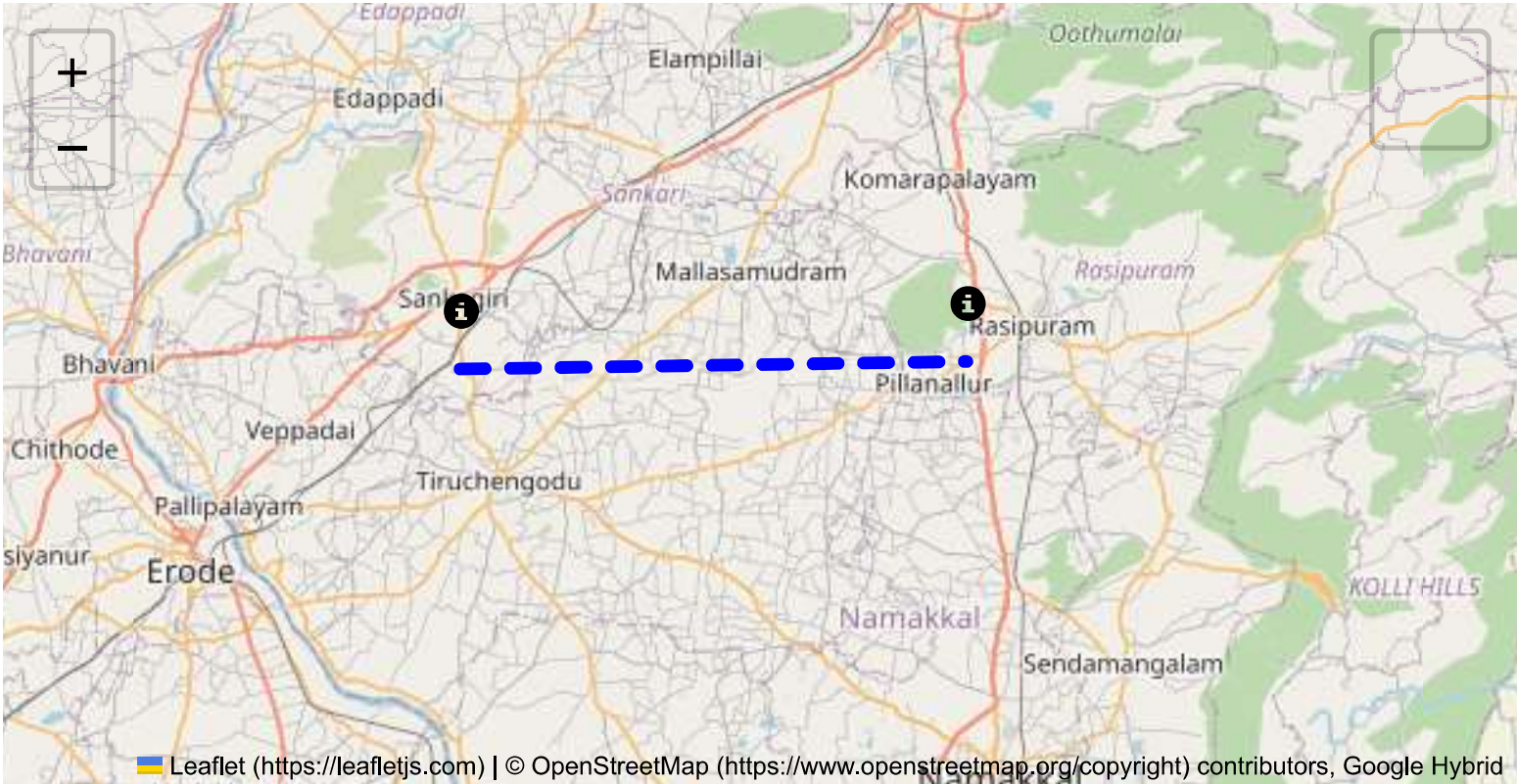
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

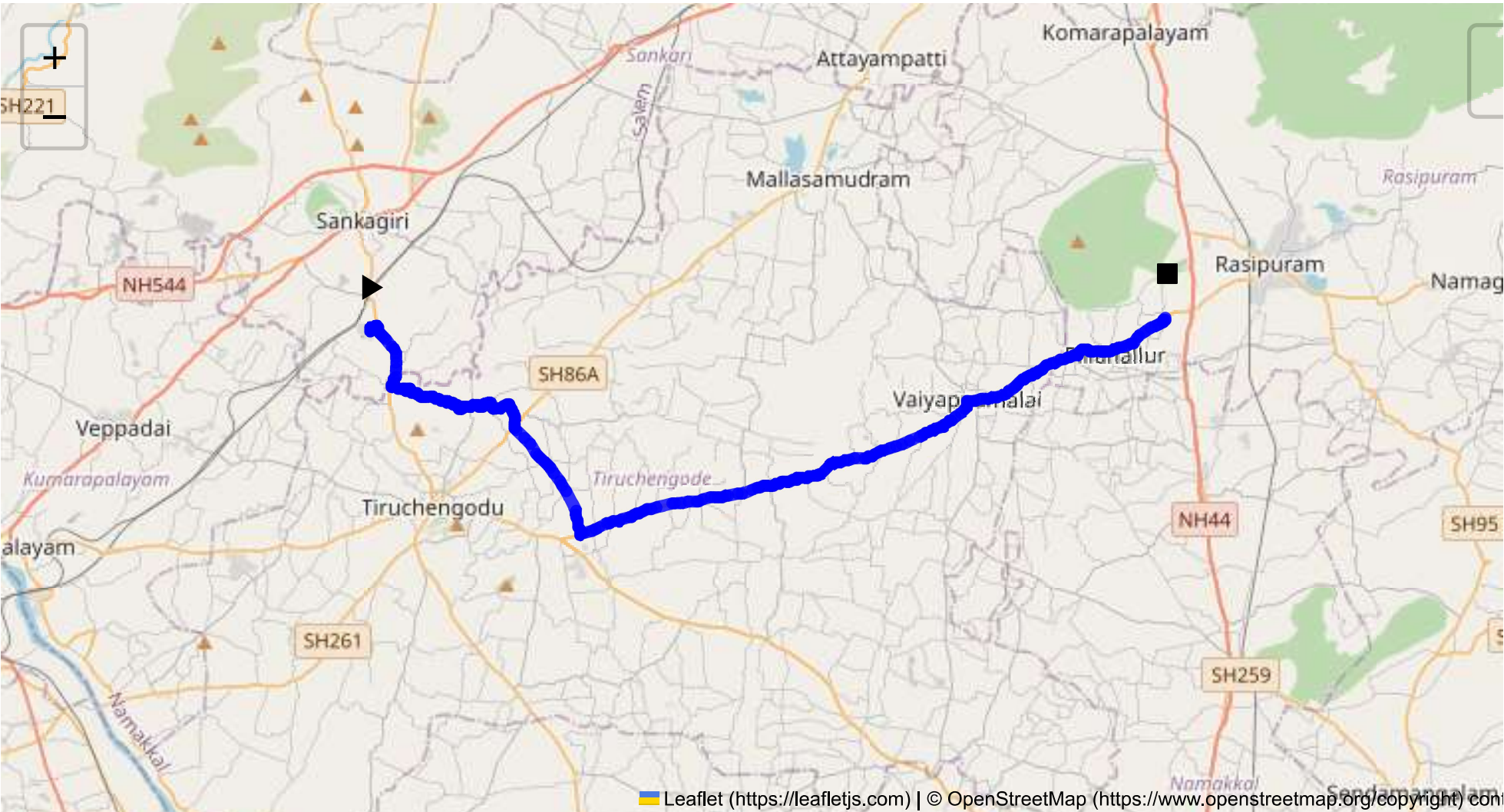
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 38.17 km
Estimated Duration: 0.9 hours
Adjusted Duration (Heavy Vehicle): 1.1 hours
Start: (11.4381, 77.8734)
End: (11.442948, 78.149474)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to C5V2+W83, Kurukkapuram in Anaipalayam spans approximately 38.17 kilometers. The journey primarily involves traveling southeastward on NH44 and SH79. The roads are mostly two-lane highways with some sections expanding into four lanes, especially closer to larger townships like Rasipuram.

Weather Conditions and Potential Hazards

The typical weather conditions in Tamil Nadu include a tropical climate, with moderate to heavy rainfall during the monsoon season (June to September) and high temperatures during the summer months (March to May). Weather-related hazards include reduced visibility and slippery roads during heavy rains, which could lead to increased stopping distances and hydroplaning risks.

Traffic Patterns

The area experiences moderate traffic, with peak hours generally occurring between 8-10 AM and 5-8 PM. Congestion is more likely near populated areas such as Rasipuram. Truck movements may also increase towards the end of the week, given local market days and supply cycles.

Road Quality and Infrastructure

The road quality on NH44 is generally good, with well-maintained surfaces and clear signs. However, SH79 may have some patches with potholes or uneven surfaces, requiring caution. Watch for local traffic and pedestrians, particularly near villages.

Alternative Routes for Emergencies

An alternative route could involve taking local roads parallel to SH79, although this may increase travel time. If NH44 is blocked, utilizing local roads towards Salem and then taking SH86 may also be an option.

Local Regulations on Hazardous Material Transport

Tamil Nadu has specific regulations affecting hazardous material transport, including mandatory permits, designated routes, and strict compliance with safety standards for loads. Drivers should ensure all documentation is in order and adhere to vehicle and speed restrictions.

Historical Incidents

There have been instances of heavy vehicle accidents in this region, primarily due to human error, poor road conditions, or weather. However, incidents involving hazardous materials are relatively rare along this specific route.

Environmental Considerations

The route passes through some agricultural and minor forest areas, which are sensitive to spills or contamination. Drivers should ensure proper containment and avoid any accidental discharge of materials.

Communication Coverage

The route generally has good mobile network coverage; however, there could be short stretches of coverage drop, especially in more rural sections. Drivers should plan for these potential dead zones, particularly for navigation and communication.

Emergency Response Times

Emergency services in urban centers like Rasipuram can respond within 15-20 minutes. However, response times in rural segments might extend up to 30-45 minutes due to distance and road conditions.

Overall Summary of Risk Assessment

The overall risk for this route includes moderate-to-high considerations due to variable road conditions and weather impacts. Traffic stress during peak hours and local regulatory compliance are critical. However, incidents specific to hazardous material transport are infrequent. Continuous monitoring of weather updates and transport regulations is advisable for a safe journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Blind Spot	Blind Spot	11.41975, 77.88075	10 KM/Hr
5	Turn	Medium	11.41876, 77.88487	30 KM/Hr
6	Turn	Medium	11.41829, 77.88568	30 KM/Hr
7	Turn	High	11.41859, 77.88692	15 KM/Hr
8	Turn	Medium	11.41852, 77.88700	30 KM/Hr
9	Turn	Medium	11.41769, 77.88730	30 KM/Hr
10	Turn	Medium	11.41630, 77.89077	30 KM/Hr
11	Turn	Medium	11.41601, 77.89198	30 KM/Hr
12	Turn	Medium	11.41603, 77.89479	30 KM/Hr
13	Turn	Medium	11.41439, 77.90025	30 KM/Hr
14	Turn	Medium	11.41377, 77.90073	30 KM/Hr
15	Turn	Medium	11.41389, 77.90161	30 KM/Hr
16	Turn	Medium	11.41325, 77.90313	30 KM/Hr
17	Turn	Medium	11.41276, 77.90349	30 KM/Hr
18	Turn	Medium	11.41227, 77.90470	30 KM/Hr
19	Turn	High	11.41230, 77.90485	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
20	Turn	Medium	11.41262, 77.90506	30 KM/Hr
21	Turn	Medium	11.41273, 77.90524	30 KM/Hr
22	Turn	High	11.41329, 77.90978	15 KM/Hr
23	Turn	Medium	11.41314, 77.91006	30 KM/Hr
24	Turn	High	11.41313, 77.91057	15 KM/Hr
25	Turn	High	11.41307, 77.91058	15 KM/Hr
26	Turn	Medium	11.41334, 77.91266	30 KM/Hr
27	Turn	High	11.41424, 77.91433	15 KM/Hr
28	Turn	Medium	11.41409, 77.91458	30 KM/Hr
29	Turn	High	11.41208, 77.91551	15 KM/Hr
30	Turn	Medium	11.41207, 77.91604	30 KM/Hr
31	Turn	High	11.41204, 77.91610	15 KM/Hr
32	Turn	Medium	11.41216, 77.91626	30 KM/Hr
33	Turn	High	11.41347, 77.92096	15 KM/Hr
34	Turn	Medium	11.41336, 77.92130	30 KM/Hr
35	Turn	High	11.40566, 77.92355	15 KM/Hr
36	Turn	High	11.40557, 77.92343	15 KM/Hr
37	Blind Spot	Blind Spot	11.36920, 77.94568	10 KM/Hr
38	Turn	Medium	11.41463, 78.08067	30 KM/Hr
39	Turn	Medium	11.43248, 78.12074	30 KM/Hr
40	Turn	Medium	11.43257, 78.12096	30 KM/Hr

Emergency Locations

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
1	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
2	school	Vidya madir school	11.4356335, 78.1381408	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.41975, 77.88075



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41876, 77.88487



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41829, 77.88568



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41859, 77.88692



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41852, 77.88700



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41769, 77.88730



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41630, 77.89077



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41601, 77.89198



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41603, 77.89479



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41439, 77.90025



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41377, 77.90073



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41389, 77.90161



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41325, 77.90313



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41276, 77.90349



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41227, 77.90470



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41230, 77.90485



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41262, 77.90506



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41273, 77.90524



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41329, 77.90978



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41314, 77.91006



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41313, 77.91057



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41307, 77.91058



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41334, 77.91266



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41424, 77.91433



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41409, 77.91458



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41208, 77.91551



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41207, 77.91604



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41204, 77.91610



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41216, 77.91626



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41347, 77.92096



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41336, 77.92130



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40566, 77.92355



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40557, 77.92343



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.36920, 77.94568



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41463, 78.08067



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43248, 78.12074



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43257, 78.12096